



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab Experiment 4

Crypt-Arithmetic Problem

Aim

To write a Python program to solve the Crypt-Arithmetic problem using Backtracking.

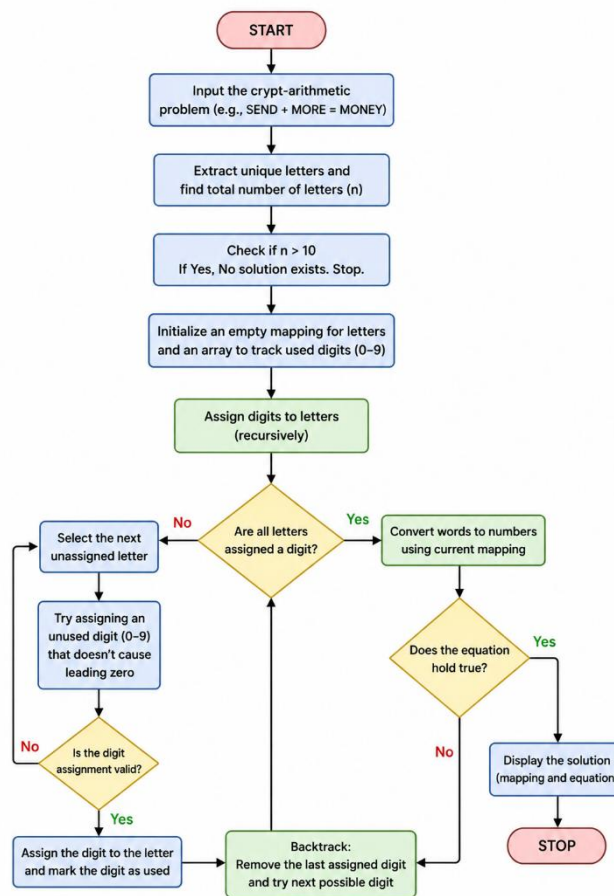
Objective

The objective of this experiment is to implement the Backtracking algorithm to solve a Crypt-Arithmetic puzzle. It aims to assign unique digits to letters such that the given arithmetic equation is satisfied.

Algorithm

1. Read the crypt-arithmetic expression.
2. Extract all unique letters from the expression.
3. Assign unique digits (0–9) to each letter using backtracking.
4. Ensure that no leading letter is assigned zero.
5. Check whether the arithmetic equation is satisfied.
6. If valid, display the solution; otherwise, backtrack and try another assignment.

Flowchart



Code

```
from itertools import permutations

letters = ('S','E','N','D','M','O','R','Y')
digits = (0,1,2,3,4,5,6,7,8,9)

for perm in permutations(digits, len(letters)):
    d = dict(zip(letters, perm))

    if d['S'] == 0 or d['M'] == 0:
        continue

    SEND = 1000*d['S'] + 100*d['E'] + 10*d['N'] + d['D']
    MORE = 1000*d['M'] + 100*d['O'] + 10*d['R'] + d['E']
```

```

MONEY = 10000*d['M'] + 1000*d['O'] + 100*d['N'] + 10*d['E'] + d['Y']
if SEND + MORE == MONEY:
    print("Solution Found:")
    print(d)
    print(f"{SEND} + {MORE} = {MONEY}")
    break

```

Output

```

Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.

===== RESTART: C:/Users/ABBIRAMY/Documents/AI-lab 4.py =====
Solution Found:
{'S': 9, 'E': 5, 'N': 6, 'D': 7, 'M': 1, 'O': 0, 'R': 8, 'Y': 2}
9567 + 1085 = 10652

```

Result

The Python program successfully solved the Crypt-Arithmetic problem by assigning unique digits to letters and satisfying the given arithmetic equation.

Conclusion

The Backtracking algorithm efficiently explores possible digit assignments and finds the correct solution while ensuring all puzzle constraints are satisfied.